Predicting Familial Versus Sporadic Neurofibromatosis Type 1 Using Clinical Symptom Data

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Abstract

Neurofibromatosis type 1 (NF1) is a common neurogenetic disorder with a highly variable clinical presentation, making symptom-based classification difficult. About half of NF1 cases are sporadic, which makes familial-versus-sporadic prediction a biologically meaningful task rather than a purely computational label split. In this project, I evaluated whether clinical symptom data alone could distinguish familial from sporadic NF1 cases. Using the UCI NF1 clinical symptom dataset, cleaned to 296 samples and 19 columns, I compared Logistic Regression, Support Vector Machine, Random Forest, and TabPFN, and then added ANOVA-based feature ranking with a top-feature Logistic Regression model. Full-feature performance was weak, with the best test-set Logistic Regression reaching accuracy 0.433, F1 0.320, and AUC 0.504. The strongest improvement came after restricting the analysis to the top 8 ANOVA-ranked symptom features, which raised Logistic Regression performance to accuracy 0.583, F1 0.419, and AUC 0.598. These results suggest that familial and sporadic NF1 are not strongly separable from symptom data alone when all symptoms are used at once, but that targeted feature selection can recover a modest and biologically interpretable signal.

Keywords: neurofibromatosis type 1; familial cases; sporadic cases; machine learning; ANOVA; feature selection

1. Introduction and Background

1.1 NF1 as a biological problem and Motivation

Neurofibromatosis type 1 (NF1) is a genetic condition that causes changes in skin pigment and tumors on nerve tissue. [1] NF1 is considered a rare disease in the general population, but it is also described as one of the most common neurogenetic disorders. [2] This makes NF1 a difficult condition to classify because symptoms can overlap and vary a lot between patients. As a result, separating familial and sporadic cases using symptom data alone is challenging. In this project, I investigated whether machine learning models could detect useful patterns in these clinical features. I began with baseline models, then added a newer method, and later used ANOVA-based feature selection to see whether a smaller set of symptoms improved performance. This variability is especially important when comparing familial and sporadic NF1. The source article notes that about 50% of NF1 cases are sporadic, arising from de novo mutations without a family history of disease. [3] Because familial and sporadic patients can still share many clinical manifestations, classification from symptoms alone is difficult.

In this report, familial' means the condition appears in a family line (suggesting inherited risk), while ‘sporadic’ refers to cases that arise without known family history, often due to de novo mutation. Even when the genetic cause is known, familial and sporadic cases can still show overlapping clinical symptoms. That overlap makes it difficult to distinguish the two groups using symptoms alone, which is exactly why this prediction task is challenging and meaningful. If ML can find patterns in symptom data, it could help researchers understand which symptoms carry the most group-level signal and which symptoms are too common in both groups to be useful for separation.

Sharafi et al. studied familial versus sporadic NF1 cases and asked whether clinical symptom data could help distinguish between the two groups. They used a retrospective dataset of 241 NF1 patients with 121 sporadic and 120 familial cases, compared symptom frequencies between the groups, used ANOVA to identify the most important distinguishing features, and then tested several machine-learning models, including k-nearest neighbors, artificial neural networks, support vector machines, decision trees, and XGBoost. Their results were

only moderately successful, with XGBoost giving the highest reported accuracy at 62.86%, and they concluded that larger, more diverse datasets would be important for improving prediction [3].
That paper motivated my project because it showed that this is a real biological classification problem, not just a random label split, and it also suggested that feature importance might matter as much as the choice of model. In my project, I followed the same general idea of using symptom data and ANOVA-based ranking, but my workflow differs in a few ways. I used a cleaned working dataset of 296 samples and 19 columns, compared to Logistic Regression, SVM, Random Forest, and TabPFN, and then retrained Logistic Regression on the top ANOVA-ranked features. So instead of only asking which model performs best, my workflow shifted toward asking which symptoms carry the most useful signal and whether a smaller, focused symptom subset improves performance more than simply adding a newer model.

1.2 Research question

Based on the biological variability of NF1 and the overlap between familial and sporadic symptom profiles, this project asked whether clinical symptom data could distinguish between the two groups. It also examined whether ANOVA-guided feature selection improved classification performance compared with using the full feature set. So, my question is:
RQ1: Can clinical symptom data distinguish familial from sporadic NF1 cases?
RQ2: Does ANOVA-based feature selection improve performance compared with using the full feature set?
RQ3: How do baseline models compare with a newer method such as TabPFN on this dataset?

2. Data Description

2.1 Dataset source and summary

For this project, I downloaded the data set from UCI Machine Learning Repository, which is the same general dataset family associated with the Sharafi et al. study. In Table 1. Dataset Summary, it shows the cleaned working version used here, the dataset contained 296 samples and 19 columns, with class counts of 161 sporadic and 135 familial cases. The target variables show the case types where 1 represents familial cases and 0 represents sporadic cases. The task was to classify classification of case type from clinical symptom variables.

Item	Value
Dataset Source	UCI NF1 Clinical Symptoms Dataset
Target Variable	Case Type (1 = Familial, 0 = Sporadic)
Samples After Cleaning	296
Features / Columns Used	19
Class Balance	Sporadic: 161 · Familial: 135

Table 1 Dataset summary: after cleaning (UCI NF1 clinical symptoms dataset).

Each row represents one NF1 case and each feature column corresponds to a clinical symptom or related clinical indicator. The target label is Case Type (1 = familial, 0 = sporadic). The cleaned dataset used in this project contains 296 cases and 19 feature columns (after removing an ID-like column), with a fairly balanced class distribution (161 sporadic, 135 familial). Some feature columns contain missing values; these were handled during preprocessing using median imputation so that models could be trained without dropping cases.

3. Methods

3.1 Overview of workflow

This project follows a structured pipeline designed to answer three related questions: whether clinical symptom features can separate familial from sporadic NF1 cases, whether selecting a smaller subset of symptoms improves prediction, and whether a newer tabular method (TabPFN) [4] [5] provides an advantage over traditional baselines. The workflow proceeds in four stages: data cleaning and preprocessing, baseline model training and evaluation using cross-validation, test-set evaluation of the best-performing approach, and interpretability-focused analysis using group summaries, ANOVA feature ranking, and permutation importance. This design keeps the analysis reproducible and makes it possible to compare models fairly using the same train/test split, the same cross-validation folds, and the same evaluation metrics.

3.2 Data loading and cleaning

The dataset was loaded from an Excel file (sheet "Dataset") and inspected for columns that do not represent clinical symptoms. An ID-like column ("Unnamed: 0") was removed when present, since this field functions as a record index rather than a symptom feature and could introduce noise or accidental leakage if the ordering of cases correlates with the target label. After removing the ID-like column, the cleaned dataset used for modeling contained 296 cases and 19 symptom-related feature columns. The target label is "Case Type," where 1 corresponds to familial cases and 0 corresponds to sporadic cases. The class distribution is moderately balanced (161 sporadic vs 135 familial), which is helpful for training classifiers but still requires attention to class-wise performance because the smaller class can be harder to detect.

3.3 Handling missing values

Clinical datasets commonly contain missing values due to incomplete charting, inconsistent reporting, or differences in clinical evaluation across patients. Dropping rows with missingness can reduce an already small sample size and may introduce bias if missingness is systematic. Therefore, missing values were handled by median imputation. Median imputation is a simple and robust strategy: it replaces missing entries in each feature with the median value of that feature in the training data. This choice avoids extreme-value distortion that can occur with mean imputation, and it allows all models to train without fail due to NaNs. Median imputation was applied within a scikit-learn pipeline so that imputation happens consistently during cross-validation without leaking information from the test fold into the training fold.

3.4 Feature scaling and preprocessing pipelines

Some machine learning models are sensitive to the scale of input features. Logistic Regression and SVM, in particular, can be influenced by feature magnitude because they rely on distances or linear decision boundaries in feature space. To avoid one feature dominating solely because it has a larger numeric range, features were standardized using z-score scaling (zero mean, unit variance) for these scale-sensitive models. In contrast, Random Forest is less sensitive to feature scaling because it is tree-based and splits on thresholds per feature. For this reason, two preprocessing pipelines were used: Scaled pipeline: median imputation + StandardScaler (used for Logistic Regression and SVM). Unscaled pipeline: median imputation only (used for Random Forest and TabPFN).

3.5 Train/test split strategy

To estimate generalization to unseen cases, the dataset was split into training and testing partitions using a stratified 80/20 split. Stratification preserves the original class balance in both the training and test sets, which reduces the risk that the test set accidentally contains mostly one class. A fixed random seed was used to make the split reproducible. The test set remained held out during cross-validation and was used only for final evaluation of selected models, which provides a more realistic estimate of out-of-sample performance.

3.6 Baseline models and "recent" model

I compared three baseline classifiers that are common for tabular clinical data: Logistic Regression (a simple linear model), SVM with an RBF kernel (a non-linear model), and Random Forest (an ensemble of decision trees). I also evaluated TabPFN as a newer tabular method designed for small datasets, to test whether a recent approach improves over the baselines.

3.7 Cross-validation and model comparison

Model comparison was performed using 5-fold stratified cross-validation on the training set. Stratified folds preserve class balance within each fold, which helps stabilize performance estimates when one class is smaller. For each fold, the model was trained on 4/5 of the training data and evaluated on the remaining 1/5. The final reported cross-validation metrics are the mean across the five folds. This approach reduces the chance that the evaluation depends too heavily on one lucky or unlucky split. It also provides a more stable estimate than a single train/validation split, which is important when n=296 is not large.

3.8 Evaluation metrics (Accuracy, F1, and AUC)

I reported accuracy (overall correctness), F1 (a balance between precision and recall), and AUC (how well the model separates the two classes across thresholds). Because the classes can overlap, AUC near 0.50 indicates performance close to random guessing.

3.9 Group-level symptom comparison (descriptive interpretability)

To better understand the dataset, I summarized average symptom values in sporadic vs familial cases and ranked features by their absolute between-group difference. This descriptive check helps confirm whether any symptoms show visible group-level differences before interpreting model results.

3.10 ANOVA feature ranking and “top feature” model

To test whether a smaller set of informative symptoms improves prediction, I used ANOVA F-scores to rank features. ANOVA measures how strongly the values of each feature differ between the two groups relative to variation within each group. Features with higher F-scores show larger group separation and are candidates for distinguishing familial vs sporadic cases. After ranking, I selected the top 8 features and retrained Logistic Regression using only those features. This creates a “top feature” model that tests the hypothesis that the full feature set may contain noise or redundant information that hurts generalization.

3.11 Permutation importance (model-based interpretability)

Finally, I used permutation importance on the best-performing model to estimate which features matter most for prediction. Permutation importance works by randomly shuffling the values of one feature in the test set and measuring how much performance drops. If shuffling a feature causes a large decrease in AUC, that feature likely aids in important predictive information. This method is model-agnostic and can be applied to different classifiers. In this project, permutation importance was used as a complementary interpretation tool alongside ANOVA ranking: ANOVA identifies features with strong group differences, while permutation importance estimates which features the trained model relies on most when making predictions.

4. Results

4.1 RQ1: Can clinical symptom data distinguish familial from sporadic NF1 cases?

Model	CV Accuracy	CV F1	CV AUC
LogReg	0.483	0.445	0.492
SVM	0.445	0.414	0.488
RandomForest	0.462	0.357	0.435
TabPFN	0.500	0.0612	0.420

Table 2. Full-feature model comparison using 5-fold cross-validation on the training split.

Using all symptom features, the models performed close to chance. Logistic Regression was the strongest baseline (CV AUC 0.492; CV F1 0.445), SVM was similar (CV AUC 0.488; CV F1 0.414), and Random Forest was lower (CV AUC 0.435; CV F1 0.357). TabPFN did not improve over the baselines (CV AUC 0.420; very low F1), suggesting weak separability in the full feature set.

4.2 RQ2: Does ANOVA-based feature selection improve performance compared with using the full feature set?

Model	Test Accuracy	Test F1	Test AUC
Full-feature Logistic Regression	0.433	0.320	0.504
Top-feature Logistic Regression	0.583	0.419	0.598

Table 3. Held-out test performance of Logistic Regression using all symptoms versus the top 8 ANOVA-ranked symptoms (higher is better).

After selecting the top 8 ANOVA-ranked symptoms, Logistic Regression improved on the held-out test set (Accuracy 0.583 vs 0.433; F1 0.419 vs 0.320; AUC 0.598 vs 0.504). This suggests the useful signal is concentrated in a smaller subset of symptoms.

RANK	FEATURE	ANOVA F-SCORE / NOTE
1	Lisch Nodules	3.184
2	Optic Glioma	2.125
3	Café au lait (CLS)	1.790
4	Skeletal Dysplasia	1.764
5	Hypertension	1.341
6	Tumor Case	0.878
7	Hamartoma	0.785
8	Plexiform Neurofibromins	0.602

Table 4. Top 8 symptom features ranked by ANOVA F-score (higher indicates larger between-group differences).

Table 4 lists the symptoms with the largest statistical differences between groups in this dataset, providing an interpretable summary of the features used in the top-feature model.

4.3 RQ3: How do baseline models compare with a newer method such as TabPFN on this dataset?

TabPFN did not outperform the baselines in cross-validation (lower AUC than Logistic Regression and very low F1). This suggests that model novelty alone does not overcome weak separability in the symptom features.

5. Comparison with Alternative Approaches

5.1 Baselines versus TabPFN

TabPFN is a recent method for tabular prediction, especially for small datasets [4]. On this dataset, it did not improve over Logistic Regression or SVM: its AUC was lower and its F1 was very low, which suggests it struggled to identify familial cases reliably. This reinforces a practical point in biomedical ML: when the underlying signal is weak, newer models do not automatically perform better.

5.2 Full features versus ANOVA-selected features

The most meaningful improvement in this project comes from comparing the full symptom feature set to an ANOVA-selected subset of symptoms. Using all features at once produced near-chance separability, which suggests that many symptoms either overlap between familial and sporadic cases or introduce noise that makes learning harder. After ranking symptoms using ANOVA and training Logistic Regression using only the top 8 features, test performance increased substantially (Accuracy 0.583 vs 0.433; F1 0.419 vs 0.320; AUC 0.598 vs 0.504). This improvement suggests that the dataset does contain useful information, but the signal is concentrated in a smaller subset of symptoms rather than spread across all features. A practical takeaway is that feature selection (or feature filtering) can be more valuable than switching model families when working with symptom-based biomedical datasets that are noisy or clinically heterogeneous.

5.3 Comparison with the source article

Sharafi et al. also report that symptom-based prediction for familial versus sporadic NF1 is only moderately successful, with their best-performing model (XGBoost) reaching 62.86% accuracy [3]. Although my results are not directly comparable due to differences in preprocessing, model selection, and evaluation setup, both studies lead to a consistent high-level conclusion: symptom overlap and variability limit how well familial and sporadic cases can be separated using symptoms alone. Importantly, Sharafi et al. emphasize that larger and more diverse datasets would improve prediction reliability, which matches what is suggested by the near-chance performance observed here when using the full feature set. In that sense, this project serves as a replication-style baseline that supports the broader claim that symptom-only signals are weak overall, but feature selection can help recover modest, interpretable structure when present.

6. Conclusion and Discussion

6.1 Main findings

Across baseline models trained on the full symptom feature set, performance remained close to chance, which indicates that familial and sporadic NF1 cases are difficult to separate using symptoms alone when all symptoms are included simultaneously. However, ANOVA-guided feature selection improved performance noticeably, showing that the dataset is not purely noise: some symptoms differ more consistently between groups than others. Taken together, the results suggest a "weak-signal" setting: full-feature learning struggles because the data contains overlap and variability, but a reduced subset of features can improve generalization and provide a clearer interpretation of what the model is using.

6.2 Biological interpretation

A plausible biological interpretation is that NF1 has high clinical variability, and that familial and sporadic cases can share many overlapping symptoms even when the underlying genetic mechanism differs. That overlap reduces separability when the model is given a wide set of symptoms, especially if many features are common across both groups and if they are inconsistently recorded. The fact that ANOVA-selected features improved performance suggests that certain symptoms appear at different rates between groups in this dataset, even if the overall boundary remains weak. This interpretation aligns with the source study's conclusion that symptom-based prediction is only moderately successful, and that richer clinical/genetic context or larger datasets may be needed for stronger discrimination.

6.3 Limitations

This project has several limitations that are likely to contribute to the near-chance full-feature performance. First, the dataset is relatively small for machine learning and may not capture the full diversity of NF1 presentations, which can reduce the stability of model estimates. Second, symptom encoding may be coarse (binary or sparse indicators), and the dataset may not include important clinical context (age, severity, genetic confirmation, care setting) that could increase separability. Third, only a small set of models and a single feature-ranking strategy (ANOVA) were explored, so there may be alternative modeling or feature-selection approaches that improve results. These limitations support the idea that symptom-only classification is inherently difficult and sensitive to dataset quality and representativeness.

6.4 Future work / remaining work

Future work could strengthen conclusions in two ways: improving modeling rigor and improving data richness. On the modeling side, one next step would be careful hyperparameter tuning for the strongest baseline (Logistic Regression and/or SVM), along with repeated train/test splits to check stability of performance. It would also be reasonable to compare ANOVA selection with other feature-selection methods (e.g., mutual information or embedded model-based selection) to verify whether the same symptoms remain important. On the data side, the most impactful improvement would be expanding the dataset and incorporating additional clinical or genetic variables, which could increase separability beyond what symptoms alone allow.

7. Data and Code Availability

The dataset is publicly available from the UCI Machine Learning Repository (NF1 clinical symptoms dataset). For reproducibility, the submission includes the dataset file, scripts, and saved outputs needed to reproduce the reported tables.

References

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